AQ3.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

If $(1,2,3)+v=(3,2,1)$ in the vector space R^3 with usual addition and scalar multiplication, where $v=(a,b,c)$, then find the value of $a+b+c$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

Which of the following statements are related to the cancellation law for vectors x,y,z in a vector space.

Statement P: $x+y=z+y \implies x=z$.
Statement Q: $x+y=z+y \implies x=y$.
Statement R: $x+y=z+y \implies y=z$.
Statement S: $x+y+z=x \implies y+z=0$

Find the number of correct statements

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2

1 point

1 point

Which of the following options is/are true for a non zero vector space V ?

- [ ] $x+y=z+y \implies x=z \ \forall x,y,z \in V$.
- [ ] $ax=bx, \ \forall\ x \in V \implies a=b, \text{ where } a,b \in R$.
- [ ] $ax=ay, \ \forall\ a \in R \implies x=y, \text{ where } x,y \in V$.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$x+y=z+y \implies x=z \ \forall x,y,z \in V$.
$ax=bx, \ \forall\ x \in V \implies a=b, \text{ where } a,b \in R$.
$ax=ay, \ \forall\ a \in R \implies x=y, \text{ where } x,y \in V$.

1 point

Which of the following options is/are true for a non zero vector space V ?

- [ ] $av=0, \ \forall\ v \in V \implies a=0$.
- [ ]

$av = 0, \forall\, a \in R \implies v = 0$.

☐ $c \in R$ and $v, w \in V$ such that, $c(v - w) \neq cv - cw$.

☐ $c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$av = 0, \forall\, v \in V \implies a = 0$.
$av = 0, \forall\, a \in R \implies v = 0$.
$c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$

Let $(3, 2, -3) \in R^3$ be a vector from the vector space R^3 with usual addition and scalar multiplication. If $c(3, 2, -3) = (-6, -4, 6)$, find the value of c.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -2
1 point
1 point
Which of the following options is/are true?
☐ The zero vector is a unique vector in a vector space.
☐ Inverse of a vector is a unique vector in a vector space.
☐ $0v \neq 0$, where v is a vector from a vector space of R.
☐ $c0 = 0$, where $c \in R$ and 0 is the zero vector from a vector space over R.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The zero vector is a unique vector in a vector space.
Inverse of a vector is a unique vector in a vector space.
$c0 = 0$, where $c \in R$ and 0 is the zero vector from a vector space over R.

Level 2

Let V be a plane plane parallel to the XY-plane. Any plane parallel to XY-plane is given by $z = c$. We define addition of two vectors $v_1 = (x_1, y_1, c)$ and $v_2 = (x_2, y_2, c)$ on V as follows: First project v_1 and v_2 on the XY-plane (we will get the vectors $(x_1, y_1, 0)$ and $(x_2, y_2, 0)$ by projection on the XY-plane) and then calculate the addition of the vectors we obtained by the projection on the XY-plane (we will obtain $(x_1 + x_2, y_1 + y_2, 0)$). Then project the obtained vector back to the plane V (we will obtain the vector $(x_1 + x_2, y_1 + y_2, c)$).
Let V be a plane which is parallel to the XY-plane, defined by $z = 2$. Let $v_1 = (1, 2, 2)$

$= (1, 2, 2)$ and $v_2 = (0, 3, 2)$ be in V. Answer questions 7 and 8.

If $v_1' = (a_1, b_1, c_1)$ and $v_2' = (a_2, b_2, c_2)$ are the projections of v_1 and v_2 on the XY - plane, respectively, then find the value of $2(a_1 + b_1 + c_1) - (a_2 + b_2 + c_2)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

If $v_1 + v_2 = (a_1, b_1, c_1)$ as per the addition defined above, then find the value of $(2a_1 + b_1 - 2c_1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Let V be a plane plane parallel to the XY -plane. Any plane parallel to the XY -plane is given by $z = c$. We define scalar multiplication of a vector $v = (x, y, c)$ on V as follows: First project v on the XY -plane first (we will get the vector $(x, y, 0)$ by projection on the XY -plane) and then calculate the scalar (α) multiple of the vector we obtained by the projection on the XY -plane (we will obtain $(\alpha x, \alpha y, 0)$). Then project the obtained vector back to the plane V (we will obtain the vector $(\alpha x, \alpha y, c)$).
Let V be a plane which is parallel to the XY -plane, defined by $z = 2$. Let $v = (1, 2, 2)$ be in V. Answer questions 9 and 10.

If $v' = (a_1, b_1, c_1)$ is the projection of v on the XY - plane, then find the value of $(a_1 + 2b_1 + c_1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5
1 point

If $4v = (a_1, b_1, c_1)$ as per the scalar multiplication defined above, then find the value of $(a_1 + 2b_1 + c_1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 22
1 point

Check Answers
Your score is: 0/10